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Department of Mathematics
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Math 101 Final Exam Term 222

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 120 minutes

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1. The global minimum value of $f(x) = (x-3)\sqrt{x}$ is at the point

(a) $(0, 0)$

(b) $(4, 2)$

(c) $\left(\frac{1}{2}, \frac{-5}{2\sqrt{2}}\right)$

(d) $(3, 0)$

(e) $(1, -2)$

2. One statement is False about $f(x) = \tanh x$

(a) $f(\ln x) = \dfrac{x^2+1}{x^2-1}$

(b) $f(x)$ is increasing on its domain

(c) $f(x)$ is an odd function passing through the origin

(d) $f'(x) = \operatorname{sech}^2 x$

(e) $f(x)$ has 2 horizontal asymptotes

3. $\lim_{x\to 0}\left(\frac{\sinh x - x}{x^3}\right) =$

(a) 1

(b) $\dfrac{1}{2}$

(c) $\dfrac{2}{3}$

(d) $\dfrac{1}{6}$

(e) $\dfrac{1}{3}$

4. The slope of the tangent line to the graph $y^2 + \ln xy = 2$ at the point $(e, 1)$ is

(a) $\dfrac{1}{e}$

(b) $\dfrac{-3}{e}$

(c) $\dfrac{-1}{3e}$

(d) $\dfrac{-1}{e}$

(e) $\dfrac{e}{3}$

5. If the constant $a > 0$ such that the function

$$f(x) = \begin{cases} \dfrac{x^3 - a^3}{x - a}, & x \neq a \\ 12, & x = a \end{cases}$$

is continuous over $(-\infty, \infty)$, then $a^2 + a =$

(a) 0

(b) 8

(c) 2

(d) 6

(e) 4

6. The number of vertical asymptotes of $f(x) = \dfrac{\ln(x^2 - 4)}{(x - 1)(x - 3)}$ is

(a) 3

(b) 4

(c) 1

(d) 0

(e) 2

7. If $\tan^{-1}(\sqrt{x + y}) = 1$, then $\dfrac{dy}{dx} =$

(a) -1

(b) 1

(c) -2

(d) 2

(e) 0

8. $\lim_{x\to 1^-}(2-x)^{\tan\left(\frac{\pi}{2}x\right)} =$

(a) $e^{-\sqrt{2}}$

(b) $e^{-\pi}$

(c) $e^{2/\pi}$

(d) $e^{1/2}$

9. If $\lim_{x\to 1}\left(\dfrac{2x-2x^2}{1-x}\right) = 2$, then the largest value of δ such that $|f(x)-2| < \epsilon$ whenever $0 < |x-1| < \delta$ is

(a) 4ϵ

(b) 2ϵ

(c) $\dfrac{\epsilon}{4}$

(d) $\dfrac{\epsilon}{2}$

(e) ϵ

10. Starting with $x_0 = 0$, the second approximation x_2 to the root of the equation $x^3+x+1=0$ is

(a) $-\dfrac{3}{4}$

(b) -1

(c) 0

(d) $-\dfrac{5}{4}$

(e) $-\dfrac{1}{2}$

11. A ladder 5 meters long is leaning against the wall of a house. The base of the ladder is pulled away from the wall at rate $\dfrac{1}{2}$ meter per second. The rate at which the angle between the ladder and wall is changing when the base of the ladder is 4 meters from the wall is:

(a) $\dfrac{1}{6}$ rad/sec

(b) $\dfrac{1}{9}$ rad/sec

(c) $\dfrac{1}{3}$ rad/sec

(d) $\dfrac{1}{4}$ rad/sec

(e) $\dfrac{1}{2}$ rad/sec

12. The minimum vertical distance between the parabolas $y = x^2 + 1$ and $y = x - x^2$ is

(a) $\frac{9}{8}$

(b) $\frac{3}{8}$

(c) $\frac{7}{8}$

(d) $\frac{1}{8}$

(e) $\frac{5}{8}$

13. $\lim_{x\to 2^-} \left(\left\lfloor \frac{x}{2} \right\rfloor - \frac{x}{2}\right) =$ (where $\lfloor x \rfloor$ is the greatest integer function)

(a) 1

(b) -1

(c) does not exist

(d) 0

(e) 2

14. Let $g(x) = \dfrac{x(x^2+1)^2}{\sqrt{2x^3-1}}$, then $g'(2) =$

(a) $\frac{13\sqrt{15}}{3}$

(b) $\frac{\sqrt{5}}{3}$

(c) $\sqrt{5}$

(d) $\frac{5\sqrt{15}}{3}$

(e) $\frac{13\sqrt{15}}{6}$

15. $\int \left(\sqrt{x} + \frac{1}{\sqrt{x}}\right)^2 dx =$

(a) $\frac{1}{2}x^2 + \ln x + C$

(b) $\frac{2}{3}x^{3/2} + 2\sqrt{x} + C$

(c) $\frac{2}{3}\sqrt{x}(1 + \ln x) + C$

(d) $\frac{1}{2}x^2 + 2x + \ln x + C$

(e) $\frac{1}{2}x - x\sqrt{x} + C$

16. Given $g(x) = \begin{cases} x^2 \sin \frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$, then

(a) $\lim_{x \to 0} g(x) = 1$

(b) g is not differentiable at $x = 0$

(c) g is differentiable at $x = 0$ and $g'(0) = 1$

(d) g is not continuous at $x = 0$

(e) g is differentiable at $x = 0$ and $g'(0) = 0$

17. One statement is False about $f(x) = \dfrac{\sin x}{x}$

(a) $\lim_{x \to 0} f(x) = 1$

(b) $f(x)$ has no vertical asymptote

(c) The maximum value of $f(x)$ is 1

(d) $f(x)$ is an even function

(e) $y = 0$ is a horizontal asymptote for $f(x)$

18. The radius r of a circle increases from $10\,m$ to $10.1\,m$. Using differential, the estimate of the increase of the area of the circle is

(a) $101\,\pi\,m^2$

(b) $100.1\pi\,m^2$

(c) $100.5\pi\,m^2$

(d) $102\,\pi\,m^2$

(e) $100\,\pi\,m^2$

19. If the inflection points for $f(x) = \dfrac{(x+1)^2}{1+x^2}$ happen at $x = \pm\sqrt{b}$, then the possible value of b is

(a) $\dfrac{1}{2}$

(b) 1

(c) $\dfrac{1}{3}$

(d) 2

(e) 3

20. Let $p(x) = g(x)\,(f(2x))^2$, $g'(1) = 0$, $g(1) = 2$, $f(2) = 2$, $f'(2) = 3$, then $p'(1) =$

(a) 12

(b) 24

(c) 32

(d) 48

(e) 46

21. $\lim_{x\to\infty} x \sin\left(\frac{1}{x}\right) =$

(a) -1

(b) 0

(c) $+\infty$

(d) 1

(e) $-\infty$

22. The range of $f(x) = \dfrac{\cos x}{2 + \sin x}$ is

(a) $\left[\dfrac{-\sqrt{2}}{2}, \dfrac{\sqrt{2}}{2}\right]$

(b) $\left[\dfrac{-\sqrt{3}}{3}, \dfrac{\sqrt{3}}{3}\right]$

(c) $[0, 2]$

(d) $\left[\dfrac{-1}{2}, \dfrac{1}{2}\right]$

(e) $[-1, 2]$

23. $\lim_{x\to 1} \frac{\sin(x-1)}{x^2-1} =$

(a) $\dfrac{1}{2}$

(b) ∞

(c) 0

(d) $-\infty$

(e) 1

24. The graph of $f(x) = e^{2/x}$ is concaving downward over the interval:

(a) $(-\infty, 2)$

(b) $(-\infty, \sqrt{e})$

(c) $\left(-\infty, \frac{1}{2}\right)$

(d) $(-\infty, e)$

(e) $(-\infty, -1)$

25. Which one of the following statements is true about $g(x) = \dfrac{x-1}{|x-1|}$,

(a) $\lim_{x\to 1} g(x) = 1$

(b) $\lim_{x\to 1} g(x)$ does not exist

(c) $\lim_{x\to -1} g(x) = -\infty$

(d) $\lim_{x\to 1} g(x) = 0$

(e) $x = 1$ is a vertical asymptote

26. $\lim_{x\to\infty} \left(\frac{1}{2x+\cos x}\right)$

(a) equals 0

(b) equals 1

(c) limit cannot be evaluated

(d) limit does not exist

(e) equals $\dfrac{2}{\pi}$

27. $\lim_{x\to 0} \frac{2-2e^{-x}}{e^x-1} =$

(a) $-\infty$

(b) 0

(c) ∞

(d) 2

(e) 1

28. $\lim_{x\to-\infty}\left(2-\frac{x+x^3}{1+x+x^2-3x^3}\right)$ is

(a) 2

(b) $\frac{1}{3}$

(c) $\frac{5}{3}$

(d) $\frac{7}{3}$

(e) $\frac{-1}{3}$

Answer Key

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Q	Ans	Brief explanation
1	**(e)**	$f' = \frac{3(x-1)}{2\sqrt{x}}$, minimum at $x = 1$: the point $(1, -2)$.
2	**(a)**	$\tanh(\ln x) = \frac{x^2-1}{x^2+1}$, not $\frac{x^2+1}{x^2-1}$, so that statement is false.
3	**(d)**	Three applications of L'Hopital give $\frac{\cosh x}{6} \to \frac{1}{6}$.
4	**(c)**	$2yy' + \frac{1}{x} + \frac{y'}{y} = 0$ at $(e, 1)$ gives $y' = -\frac{1}{3e}$.
5	**(d)**	$3a^2 = 12 \Rightarrow a = 2$, so $a^2 + a = 6$.
6	**(a)**	Domain $\lvert x\rvert > 2$: asymptotes at $x = -2$, 2, 3 ($x = 1$ is not in the domain).
7	**(a)**	$x + y = \tan^2 1$ is constant, so $y' = -1$.
8	**(c)**	$\ln y = \frac{\ln(2-x)}{\cot(\pi x/2)} \to \frac{2}{\pi}$, so the limit is $e^{2/\pi}$.
9	**(d)**	$f(x) = 2x$ for $x \neq 1$, so $\lvert f - 2\rvert = 2\lvert x - 1\rvert < \epsilon$ gives $\delta = \epsilon/2$.
10	**(a)**	$x_1 = -1$, $x_2 = -1 + \frac{1}{4} = -\frac{3}{4}$.
11	**(a)**	$\cos\theta\, \theta' = \frac{1}{5}x'$ with $\cos\theta = \frac{3}{5}$: $\theta' = \frac{1}{6}$ rad/s.
12	**(c)**	$D = 2x^2 - x + 1$, $D' = 0$ at $x = \frac{1}{4}$, $D_{\min} = \frac{7}{8}$.
13	**(b)**	$\lfloor x/2 \rfloor \to 0$ and $x/2 \to 1$ from the left, so the limit is -1.
14	**(a)**	Log differentiation: $g(2) = \frac{50}{\sqrt{15}}$, $g'/g = \frac{13}{10}$, so $g'(2) = \frac{13\sqrt{15}}{3}$.

Q	Ans	Brief explanation
15	**(d)**	Expand: $x + 2 + \frac{1}{x}$, so the integral is $\frac{1}{2}x^2 + 2x + \ln x + C$.
16	**(e)**	$g'(0) = \lim h \sin \frac{1}{h} = 0$ by squeeze, so differentiable with $g'(0) = 0$.
17	**(c)**	$f(x) < 1$ for all $x \neq 0$ and $f(0)$ is undefined, so 1 is not attained: false.
18	**(d)**	$dA = 2\pi r\, dr = 2\pi$; estimated area $100\pi + 2\pi = 102\pi$ (official key).
19	**(e)**	$f = 1 + \frac{2x}{1+x^2}$, $f'' = \frac{4x(x^2-3)}{(1+x^2)^3}$: zeros $\pm\sqrt{3}$, so $b = 3$.
20	**(d)**	$p' = g'f^2 + 4gff'$ gives $0 + 4 \cdot 2 \cdot 2 \cdot 3 = 48$.
21	**(d)**	With $u = 1/x$: $\frac{\sin u}{u} \to 1$.
22	**(b)**	$f' = 0 \Rightarrow \sin x = -\frac{1}{2}$, $f = \pm\frac{\sqrt{3}/2}{3/2} = \pm\frac{\sqrt{3}}{3}$.
23	**(a)**	$\frac{\sin(x-1)}{x-1} \cdot \frac{1}{x+1} \to 1 \cdot \frac{1}{2} = \frac{1}{2}$.
24	**(e)**	$f'' = \frac{4(x+1)}{x^4}e^{2/x} < 0$ exactly for $x < -1$.
25	**(b)**	One-sided limits are 1 and -1, so the limit does not exist.
26	**(a)**	$0 < \frac{1}{2x+\cos x} \leq \frac{1}{2x-1} \to 0$, so the limit is 0.
27	**(d)**	L'Hopital: $\frac{2e^{-x}}{e^x} \to 2$.
28	**(d)**	Fraction $\to \frac{1}{-3}$, so the limit is $2 + \frac{1}{3} = \frac{7}{3}$.